Full Text, Language, and Embeddings

Full Text, Language, and Embeddings I

Beyond bibliographic metadata, the pipeline mines the textual content of each record. This stage bridges the gap between a bibliographic inventory and a semantic understanding of the literature.

Full-Text Availability I

An open-access resolver maps each record to a downloadable PDF where one exists (a known `pdf_url`, or an Unpaywall lookup by DOI), and an opt-in, network-gated downloader fetches it to a deterministic path. Full-text availability is summarized without requiring any download, so the offline default still reports coverage. For this run:

- **Abstract coverage**: {{ABSTRACT_COVERAGE_PCT}}% of records ({{ABSTRACT_COUNT}} of {{CORPUS_SIZE}}) carry an abstract; {{NO_ABSTRACT_COUNT}} records lack one.
- **Open-access status**: {{OA_PCT}}% of records are open access ({{OA_COUNT}} records); the remainder are closed or unknown.

Full-Text Availability II

- **PDF availability**: {{PDF_AVAIL_PCT}}% of records ({{PDF_AVAIL_COUNT}}) have a direct PDF link; {{PUBLISHER_PDF_COUNT}} have a publisher PDF, and {{NO_FULLTEXT_COUNT}} have no full-text source available.

The identifier coverage for this corpus is: {{DOI_COUNT}} DOIs, {{OPENALEX_ID_COUNT}} OpenAlex IDs, and {{ARXIV_ID_COUNT}} arXiv IDs. DOI coverage dominates and supports cross-engine de-duplication.

Language and Entity Extraction I

Titles, abstracts, and (when present) full text are tokenized and reduced to keyphrases and named entities by offline, dependency-light extractors — no mandatory LLM.

Term-frequency statistics drive a TF-IDF representation over a {{NUM_VOCAB_FEATURES}}-feature vocabulary. The most frequent terms in the corpus are: {{TOP_VOCAB_TERMS}}. These terms reflect the configured domain and the records retained by the retrieval and filtering policy; fixture-derived terms are not evidence about the live field.

Embeddings I

Every title, abstract, and full text is embedded into a shared vector space by a deterministic, offline method — TF-IDF followed by truncated SVD, i.e. latent semantic analysis [@deerwester1990indexing]. The embedding dimensionality is 50 components (configurable via `project_config.embeddings.n_components`), and the TF-IDF vocabulary is capped at {{NUM_VOCAB_FEATURES}} features (configurable via `project_config.embeddings.max_features`). The embedding is byte-stable across runs: the same input text always yields identical vectors, so the derived similarity matrix, nearest-neighbour lists, clusters, and two-dimensional projection are all reproducible.

An optional transformer backend can be enabled by setting `project_config.embeddings.method: transformer` (requires the `embeddings` extra), which upgrades the embedding fidelity without changing the interface or downstream analysis.

Embeddings II

The embeddings support semantic retrieval over the corpus and feed two visualizations: a PCA two-dimensional projection ((Figure pca embeddings)) that maps the topical geography of the literature, and a hierarchical clustering dendrogram ((Figure dendrogram)) that reveals the similarity structure of the document collection.